PRIYA DHARSHINI A
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CAREER OBJECTIVE

Final-year B.Tech Biotechnology student who builds Python and Flask web applications for clinical and genomic data, backed by hands-on bioinformatics and microbiology lab experience. Looking for gaining Interdisciplinary research experience, Hands-on exposure to advanced instruments and research tools and Mentorship from experienced researchers.

EDUCATION

Bachelor of Technology in Biotechnology (2023 – 2027)
Dhanalakshmi Srinivasan University, Trichy | CGPA: 8.22

TECHNICAL SKILLS

- **Software Development:** Python, Flask, SQL/SQLite, REST APIs, HTML/CSS, JavaScript, Git/GitHub
- **Data & Visualization:** Pandas, ggplot2, matplotlib, ReportLab (PDF generation), py3Dmol (3D structure rendering)
- **Bioinformatics Tools & Databases:** BioPython, NCBI, UniProt, BLAST, dbSNP, PharmGKB, Stanford HIVdb API, Molecular Docking
- **Sequence & NGS Analysis:** Sequence alignment, phylogenetic analysis, statistical analysis
- **Wet Lab Techniques:** Microbiology, Molecular Biology, Bioprocessing Techniques

PROJECTS

HIV Antiviral Resistance Scanner | Flask Web Application

- Built a full-stack Flask web app that takes patient genomic input, queries the Stanford HIVdb Sierra GraphQL API, and auto-generates clinical resistance reports.
- Designed a clinical-style UI with patient intake forms, color-coded resistance summaries, and interactive 3D protein structure visualization (py3Dmol).

Tools: Python, Flask, BioPython, Pandas, py3Dmol, HTML/CSS, Stanford HIVdb API
Link: https://github.com/Priyadharshini182/hiv-resistance-scanner

AMR Surveillance Dashboard | Flask + SQLite Web App

- Developed a Flask/SQLite application where users upload bacterial genome files with 95% detection accuracy, backed by data accuracy checks on incoming records. (mecA, blaTEM, vanA, and others).
- Generated automated PDF clinical reports (ReportLab) and hospital-wide resistance trend charts (matplotlib).

Tools: Python, Flask, SQLite, ReportLab, matplotlib, Pandas

AMR Detection Pipelin | Data Analysis Pipeline

- Built an end-to-end AMR detection pipeline for bacterial clinical isolates across 3 species and 3 antibiotics, classifying isolates via EUCAST/CLSI MIC breakpoints.
- Automated the workflow from raw MIC data to publication-ready antibiogram heatmaps, cutting manual analysis time.

Tools: Python, Pandas, R, ggplot | GitHub: github.com/Priyadharshini182/AMR-Detection-pipeline

EXPERIENCE

Bioinformatics Basics to Advance (Virtual internship**)** | BioGrademy | June – July 2026

- Retrieved sequences from NCBI and conducted BLAST-based homology analysis; constructed multiple sequence alignments (ClustalW) and phylogenetic trees.
- Executed molecular docking (PyRx/CB-Dock) and mapped protein-protein interaction networks using Cytoscape.

Bioprocess Technology | Evolute Bioscience Pvt Ltd | June – July 2026

- Conducted shake flask fermentation studies and quantified biomolecule yield using UV-Vis spectrophotometry; maintained SOP-compliant documentation.

Microbiology and Molecular Techniques | TriBiotech Pvt Ltd | March – April 2025

- Conducted antimicrobial susceptibility testing (disc diffusion, MIC) applying EUCAST/CLSI breakpoints; performed DNA/RNA extraction, PCR, qPCR, and gel electrophoresis.

CERTIFICATIONS

- Basics to Advanced Bioinformatics — BioGrademy

- Bioinformatics Data Analysis: Master Python, Linux & R — Udemy